WE DEVELOP SOFTWARE LLC

ChristEye Living Table & Energy Framework

A documented research concept: particles, neural systems, biological production, and energy accounting

Prepared for Brad Gearon | November 2, 2026 | Version 0.5

Purpose and scope

This report consolidates the ChristEye ideas discussed in this conversation, emphasizing the living table, neuron and axon structure, learning efficiency, nuclear-to-biological energy pathways, and material outputs. It also records the related medicine, cancer, computing, animal-communication, and materials concepts. It is a research framework, not a validated device, medical treatment, nuclear apparatus design, or experimental protocol.

Evidence labels used throughout

Label	Meaning
Established	A physical principle or biological mechanism supported by the cited literature; not proof of ChristEye as an integrated system.
Proposed	A ChristEye design choice, analogy, metric, or hypothesis requiring testing.
Illustrative	A worked calculation using explicitly chosen assumptions; not measured performance.
Unknown	A necessary parameter or capability not established by the available evidence.

Faith and scientific terminology

Brad uses Soul for an organized system and the continuation of life; Winds for active signals or flows; and “soul death” for harmful loss of coordination. These definitions document his philosophical framework. No conversion constant, conserved soul quantity, angelic authentication, or spiritual energy mechanism has been measured. The scientific model therefore uses atoms, charge, energy, information, biological function, and viability as separate variables.

Earlier conversational probabilities, trillion-dollar scenarios, and assurances of feasibility are not experimental results and are not adopted as scientific estimates here. No performance claim follows merely from naming a component RAM, CPU, Wind, or Soul.

1 / Framework and system boundaries

ChristEye term	Operational interpretation	Limit
Soul	Integration and continuity of an organized system	Philosophical label, not an SI physical quantity
Lotus	Processing and coordinated action	Requires a defined controller and task
Marrow	Structure, inherited capabilities, cellular integrity	Not equivalent to anatomical bone marrow
Winds	Electrical signals, molecular signals, or energy flow	Different flows retain different units
Roots	Retained history, learned parameters, material resources	Memory and chemical storage are distinct
Heart	Needs, targets, homeostatic regulation	Targets must be measurable
Spirit / knowledge	Driving input or process advancement in Brad’s framework	Operational input is specified separately from energy and matter
Understanding	Ability to sustain and respond with life, in Brad’s definition	Model with measured reliability, endurance, and viability

Two pathways, one control loop

Energy pathway: an external source supplies engineered conversion equipment; regulated electricity then supplies immediate loads and a biological production system. Energy-rich products are collected and stored.

Material pathway: carbon sources, water, nitrogen, minerals, and other nutrients become specified products, biomass, wastes, and gas outputs. Reactor energy does not replace these atoms.

Control loop: sensors → software or experimental neural controller → electrical interface → supported actuator → process → sensors. Living controllers remain separated from damaging radiation and reactor heat. Axons transmit control information, not bulk nuclear power.

What the proposed “living periodic table” means

It is a functional taxonomy, not a replacement for the chemical periodic table. Electrons and protons are particles; ATP and tau are molecules/proteins; axons and mitochondria are cell structures; neurons and astrocytes are cell types. These levels should not be treated as interchangeable elements. Physical conservation laws apply across every level.

Embryos and developmental terminology

The “crown” recalled in discussion was clarified as neural crest cells. Neural crest is a developmental population with multiple derivatives, not an embryo-wide controller. Neural stem cells generate defined neural lineages. A human zygote usually begins with sperm-egg fertilization, but many organisms also reproduce asexually; development does not provide a known physical measure of soul. [R10]

2 / The provisional living table

Code / level	Component and role	Measurements / constraints
Nr / cell	Neuron: integrates inputs, generates activity, communicates through synapses	Correct responses per joule; error rate; viability; stability
Ax / structure	Axon: propagates action potentials; transports cellular cargo	Energy per delivered spike; delay; failure probability; morphology
As / cell	Astrocyte: regulates neural environment and provides metabolic support	Ion regulation; support demand; viability; context-dependent signaling
Ol / cell	Oligodendrocyte: myelin in the central nervous system	Myelination; signaling cost and reliability
Sc / cell	Schwann cell: peripheral nerve support and myelin	Axon association; conduction and repair measures
Mi / cell	Microglia: resident CNS immune surveillance	Inflammatory state; clearance; unintended injury
Ns / cell	Neural stem cell: self-renewal and specified neural differentiation	Lineage identity; differentiation yield; unwanted growth
Nc / cell	Neural crest cell: embryonic migration and diverse tissue derivatives	Developmental state; migration; lineage control
Sy / junction	Synapse: chemical or electrical communication	Transmission probability; strength; plasticity; energy cost
Mt / organelle	Mitochondrion: metabolic energy conversion and local regulation	ATP production; respiration; membrane potential
Rb / machinery	Ribosome: translates mRNA into proteins	Protein identity; synthesis rate; fidelity
Ef / module	Specified effector cells: make or release a defined product	Product identity; mass/time; purity; energy yield
St / module	Chemical storage: retains collected product	Recoverable energy/mass; capacity; degradation losses

Codes are proposed ChristEye abbreviations, not standard scientific symbols. All cells require appropriate environments and supplies. Neural support cells and developmental cells are not automatically therapeutic effectors. The first prototype need not include every category. [R1-R4, R10]

3 / Particles, ions, and neuronal machinery

Component	Physical or biological role
Electron (e−)	Negative charge: $-1.602176634 \times 10^{-19}$ C. Electron transfer matters in metabolism and electrodes. Neural tissue does not conduct mainly as a metal-like stream of free electrons.
Proton (p^+ / H^+)	Positive charge of the same magnitude. Nuclear proton number defines the element. Hydrogen-ion gradients across mitochondrial membranes help drive ATP synthesis.
Neutron (n)	Electrically neutral nuclear constituent; relevant to isotopes and nuclear reactions, not ordinary action-potential charge transport.
Na^+ and K^+	Their gradients and membrane channels support typical neuronal action potentials.
Ca^{2+} and Cl^-	Calcium participates in secretion and signaling; chloride contributes to inhibitory signaling depending on its gradient.
Membrane and ion channels	Lipid barrier acts as an electrical capacitor; selective channels provide regulated ionic conductances.
Na^+/K^+-ATPase	Consumes ATP to move 3 Na^+ outward and 2 K^+ inward per pump cycle, maintaining gradients over time.
Dendrites / soma / nucleus	Receive and integrate inputs; support metabolism and gene regulation. Most neurons retain their nucleus in the soma, not the axon.
Axon initial segment / nodes	Specialized regions supporting spike initiation and propagation; myelin modifies conduction geometry and energy cost.
Vesicles / receptors	Package and detect chemical signals. Their content depends on cellular machinery, not the semantic label applied to the cell.
Microtubules / actin / tau	Internal structure and transport organization. Tau is a protein, not a cell type. Motor proteins transport cargo using chemical energy.
mRNA / ribosomes / ER / Golgi	Specify, synthesize, process, and route proteins. Some local axonal protein synthesis exists; this does not confer arbitrary drug manufacture.

An action potential is a regenerating change in membrane voltage supported by local ion gradients. It is not one packet of reactor energy traveling intact from end to end. Axonal local synthesis and mitochondrial support are documented phenomena. [R1, R2, R9]

4 / Electrical and metabolic equations

Electrochemical driving force - established model

$$E_i = (RT / z_iF) \ln(c_i,out / c_i,in)$$

Nernst equilibrium potential E_i is in volts. $R = 8.314$ J mol^{-1} K^{-1}; T is absolute temperature; z_i is ion valence; $F \approx 96{,}485$ C/mol; c represents compatible concentration approximations (activities are needed for higher precision).

$$I_i = g_i(V_m - E_i)$$

$$C_m \, dV_m/dt = Iexternal - \Sigma \, I_i$$

This lumped membrane model uses total capacitance C_m in farads and currents in amperes, with outward ionic current positive. An extended axon needs a spatial cable model, geometry, channel kinetics, temperature, and boundary conditions. This is not a complete neuronal simulation.

$$\Delta G_i,in\rightarrow out = RT \ln(aout/ain) + z_iF(\psi out - \psi in)$$

This is molar free-energy change for moving an ion from inside to outside. It distinguishes electrical energy from concentration-gradient energy.

Axon geometry and pump budget

$$Aaxon \approx 2\pi rL \; ; \; Caxon \approx c_m \, Aaxon \; ; \; Raxial = \rho L/(\pi r^2)$$

r and L are radius and length; c_m is capacitance per area; ρ is axial resistivity. These cylindrical approximations omit branches and detailed myelin geometry. Larger radius is not simply “more efficient”: speed, surface area, volume, and maintenance change together.

$$EATP \approx nATP \, |\Delta GATP| \; ; \; npump \; cycles \approx nNa,restored / 3$$

The second relation accounts for sodium restoration by the Na/K pump, not all neuronal energy costs. ΔGATP depends on cellular conditions; do not insert a universal ATP energy without specifying them. Synaptic release, transport, protein turnover, glial support, and maintenance also consume energy. [R1, R3]

A component record - proposed

$$C_i = (m_i, P_i, R_i, \varepsilon_i, \tau_i, V_i)$$

Record mass m (kg), power P (W), useful output rate R (task-defined units/s), error fraction ε, delay τ (s), and viable fraction V. Synaptic weight is a functional connection parameter; it is not mass.

5 / Learning efficiency and sizing

$$C_{task} = E_{total} / N_{correct} \text{ [J/correct response]}$$

$$G_{training} = 1 - C_{after}/C_{before}$$

$$e_{Ax} = E_{propagation} / N_{successfully\ delivered\ spikes}$$

$$\Phi_{mass} = N_{correct} / (E_{total}\, m_{system})$$

Compare matched tasks, accuracy, durations, and environmental conditions. Report both incremental activity cost and whole-system cost. Include training energy; a controller that saves energy later may still take time to repay its training cost.

$$E_{amortized/task} = E_{training}/N_{lifetime\ tasks} + E_{operation/task}$$

Sizing must be measured, not inferred from reactor watts

$$N_n \geq R_{required} / r_{validated} \text{ [only under a tested scaling model]}$$

$$N_{connections} = N_n \bar{k}$$

$$P_c = N_n p_n + P_{support} + P_{interface} + P_{life\text{-}support}$$

Nn is neuron count; rvalidated is a measured effective decision rate per neuron-equivalent. Neural computation is distributed and generally not linearly scalable, so the first inequality is only a provisional capacity bound. Connection count is not necessarily distinct axon count. Geometry and measured cell volumes/densities are needed to estimate tissue mass.

$$m_{tissue} \approx \rho_{tissue} V_{tissue} \; ; \; V_{axons} \approx \Sigma \pi r_i^2 L_i$$

Do not double-count axons already included in tissue volume. Support cells, extracellular space, fluid channels, and packaging also occupy volume. No justified absolute neuron count, axon mass, or high-power tolerance follows from current ChristEye data.

What training evidence does and does not establish

DishBrain reported apparent learning within five minutes of simplified Pong gameplay in cultured human or mouse networks linked to electronics. This excludes culture preparation and interface construction. It does not establish general reasoning, arbitrary molecular synthesis, a matched advantage over LLMs, or fusion-power handling. [R4]

A specific mouse cortical-axon study found approximately halved propagation costs with myelination. That does not mean all axons improve their power capacity indefinitely through training. Test performance and energy together rather than assuming stronger firing is better. [R3]

6 / Nuclear energy, storage, and conservation

Separate nuclear reactions from electron chemistry

Ionization: $X + E_{ion} \rightarrow X^{+} + e^{-}$

D–T fusion: $^{2}H + {}^{3}H \rightarrow {}^{4}He + n + 17.6\ MeV$

Representative fission: heavy nucleus + n → fragments + neutrons + energy

$Q = (m_{initial} - m_{final})c^{2}$; $P_{nuclear} = R_{reaction}\, Q$

Deuterium is heavy hydrogen; tritium is another hydrogen isotope. D–T fusion releases about 2.82×10^{-12} J per event, with about 14.1 MeV in the neutron and 3.5 MeV in the helium nucleus. Representative uranium fission releases about 200 MeV, or 3.20×10^{-11} J per event. These are reaction energies, not plant net outputs. Removing an electron alone does not cause fusion. [R5, R6]

Whole-system energy allocation

$P_a = \eta_{conversion} P_{nuclear} - P_{plant}$

$P_d + P_s + P_c \leq P_a$

$dE_s/dt = \eta_s P_s - P_r/\eta_r - P_{loss}$

Pa: available electricity after plant loads; Pd: direct delivered electricity; Ps: storage input; Pc: controller and support; Es: stored chemical energy; Pr: electricity recovered from storage; ηs and ηr: storage and recovery efficiencies. The last expression assumes no uncounted energy-bearing feedstock. If such feedstocks are present, add their energy explicitly.

Illustrative allocation, not a plant forecast

Assumption	Value
Available electricity / direct use / storage / control	100 / 60 / 39 / 1 kW
Storage efficiency / recovery efficiency	40% / 40%
Charging time	10 hours
Chemical energy added, ignoring other losses	39 × 10 × 0.40 = 156 kWh
Later recoverable electricity	156 × 0.40 = 62.4 kWh

Round-trip efficiency of this assumed storage branch is 16%. A five-percentage-point storage-efficiency improvement at 39 kW adds 1.95 kW of chemical storage, equivalent to 0.78 kW of eventual electrical recovery at 40%, before further losses. Compare that gain with incremental controller overhead and its timing. Unlimited simultaneous use and storage are excluded by the energy balance.

7 / Chemical outputs and worked calculation

Established example: microbial acetate production

Microbial electrosynthesis can use electrical inputs to support conversion of CO_2 into organic products. Acetate is a documented example, not a product that neurons are known to manufacture on command. [R7]

$$2CO_2 + 7H^+ + 8e^- \rightarrow CH_3COO^- + 2H_2O$$

This is a reduction half-reaction. An accompanying oxidation reaction and complete equipment balance are necessary. Eight electron equivalents are needed per acetate ion.

$$\dot{n}_{acetate} = \eta_F I/(8F)$$

Chosen assumptions / calculation	Result
Current I; product-specific Faradaic efficiency η_F	1 A; 0.80
Acetate rate = 0.80/[8 × 96,485]	1.036×10^{-6} mol/s
Acetate produced in 86,400 s	0.08955 mol/day
Acetate-ion mass at 59.044 g/mol	5.29 g/day
CO_2 incorporated in acetate: 2 × 44.01 × 0.08955	7.88 g/day
Assumed full-cell voltage; electrical input VIt	3 V; 72 Wh/day

This is a stoichiometric illustration, not a predicted culture yield. It excludes additional carbon used for biomass and side products, and excludes pumping, cooling, life support, recovery, and maintenance. Near-100% Faradaic efficiency would not mean near-100% energy efficiency.

Products, biomass, and energy are different outputs

$$E_{stored} = m_{product} h_{recoverable}$$

$$\eta_{storage} = \Delta E_{stored}/(E_{electricity} + E_{chemical\ inputs})$$

$$dn/dt = S\,v + f_{in} - f_{out}$$

h is recoverable chemical energy per mass under a specified process. S is the stoichiometric matrix; v is reaction flux; f denotes material flows. Element balances require conservation of carbon, hydrogen, oxygen, nitrogen, phosphorus, sulfur, and other supplied elements.

Protein production needs amino-acid composition and nitrogen inputs; lipid and carbohydrate production need their own pathways. Protein content is not a proxy for energy-storage efficiency. Living biomass uses stored energy to maintain itself. A useful production plan must identify the organism, product, pathway, substrate, losses, and collection process before outputs can be predicted.

8 / Damage, repair, and medical concepts

A proposed functional-damage index

$$D(t) = w_1 d_{viability} + w_2 d_{function} + w_3 d_{genomic\text{-}integrity}$$

Each d must be defined as a normalized impairment measure, with higher values meaning greater impairment; nonnegative weights can sum to one. This is a proposed engineering index, not a validated diagnosis or soul measurement. The weights and predictive value require independent testing. A single scalar can conceal conflicting outcomes and should be accompanied by its individual measurements.

State	Required distinction
Healthy coordination	Maintain function without unnecessary stimulation
Repairable injury	Support demonstrated repair while monitoring abnormal growth
Dangerous persistence	Identify and restrain or remove cells that threaten tissue function
Irreversible damage	Avoid prolonging harmful activity; assess clearance and tissue support

ChristEye Sylph - proposed cancer surveillance

The concept is selective sensing and controlled response, not indiscriminate continuous growth. Cancer can involve genetic and epigenetic changes, abnormal proliferation, and evasion of death signals. Smoking, radiation, and certain infections act through different mechanisms; a virus does not generally arise by simply attaching to an mRNA fragment. Normal mRNA is essential to cells. [R8]

Neuronal or glial signals can promote certain tumors. Astrocytes, stem cells, and neural crest cells therefore cannot be assumed to provide cancer protection. The first test must distinguish healthy, injured, and malignant tissue and demonstrate no tumor-growth acceleration. [R11]

Adaptive medicine and a living pill

Controlled secretion by engineered cells has experimental precedents, including light-controlled therapeutic protein release in mice. That does not validate an oral mixture of neurons, its survival in the gut, brain sensing, systemic distribution, or dose control. [R12]

ADHD concepts discussed included timed delivery of existing medicine and speculative cellular drug production. Training alone does not provide amphetamine biosynthesis. No formulation, dose, manufacturing pathway, treatment efficacy, or human use is specified here. Cancer treatment and prevention likewise require separate, indication-specific evidence.

9 / Broader applications and research gates

Concept discussed	Documented status and testable next question
Biological RAM / CPU / graphics	Conceptual specialized modules. Can a defined task meet reproducible accuracy, retention, latency, and energy benchmarks?
Living workstation	Conceptual nutrient, waste, thermal, interface, and packaging systems. Can a small culture remain stable and measurable?
Diamond / new material	No new material has been characterized here. Can control improve a defined crystallization process, with composition and structure independently verified?
Food and energy reservoir	Use external energy and nutrients to make biomass or compounds. Does the full energy/material balance outperform alternatives?
Fusion / fission integration	Source-level concept only. No demonstrated ChristEye nuclear reaction control or net-positive cold fusion.
Animal communication / Astral Dragoth	Long-term concept involving cat-, dragon-, and mammoth-inspired traits and vocal communication. Start with digital simulation and non-invasive behavior; no creature or humanlike speech capability is established.
Digital companions	A software environment for interaction and learning concepts; conversational output does not establish biological life or spiritual provenance.
Provenance and blockchain	Can document records and timestamps; cannot authenticate scientific truth, spiritual authority, or ownership by itself.
Valuation / doctoral outreach	No validated probability of cure or dollar valuation. Research collaboration and doctoral study require evidence and institutional review.

Staged research plan

Gate 1: Choose one nonmedical, nonnuclear control task and one measurable reporter. Define success and failure before experiments.

Gate 2: Establish a conventional controller baseline. Characterize sensors, actuator delays, power demand, and measurement uncertainty.

Gate 3: Evaluate a neural controller under matched conditions, including untrained controls, repeated runs, learning retention, drift, and full overhead.

Gate 4: Couple to a contained biological output only after reliable signaling is demonstrated. Quantify identity, mass, energy, side products, and stopping behavior.

Gate 5: Require independent replication before stronger claims. Medical, animal-development, or nuclear extensions are separate specialist research programs, not automatic consequences.

10 / Revision: knowledge and understanding

Revision basis: two screenshots supplied by Brad on November 2, 2026. The visible explanations, rather than account identity or affiliation, are the source of the conceptual changes below. The first screenshot's obscured final sentence is completed by the overlapping second screenshot. No hidden text is inferred.

The explanation, faithfully summarized

Brad proposes that resources and processes play roles analogous to particles in a living system; that axons may develop a trainable capacity to sustain demands; and that neurons may improve input/output efficiency. He defines "knowledge" as something advancing a process and "understanding" as the ability to sustain and respond with life. At the molecular level, he analogizes electron/proton balance to understanding and propensity to bind another molecule to knowledge.

Brad attributes the vocabulary to the Bible and spiritual teaching. No specific verses are supplied in these screenshots. This report records that attribution without claiming that scripture establishes the proposed chemical or neural laws. Molecular binding does not require demonstrated intention, will, or semantic knowledge.

ChristEye expression	Operational counterpart	Required distinction
Knowledge / advancement	Input u(t), stimulus, reaction flux, or causal influence	Information, material, and energy inputs have different units
Understanding / sustain and respond	Reliable-response envelope, recovery, endurance, viability	Not literal semantic comprehension in an individual axon
Food / resources	Substrates, nutrients, oxygen where required, and metabolic supply	Food is not a replacement definition for electron or proton
Force / demand	Specify mechanical stress (Pa), electric field (V/m), current (A), or signal rate (Hz)	Do not combine these into one uncalibrated force score
Molecular balance / binding	Charge, structure, free energy, kinetic rates, environment	Charge neutrality alone does not predict affinity

New deductions worth testing

1. Resource supply and signaling demand must be modeled together: reliability may fall when demand exceeds maintenance capacity. 2. "Understanding" is better represented by a conditional performance envelope than one permanent number. 3. Learning should be compared with maturation, changing nutrients, and support-cell effects. 4. Better signaling accuracy, endurance, and lower energy cost can conflict; improvement in one does not establish improvement in all. These are testable research questions, not discoveries established by the screenshots.

An axon is part of a neuron; the brainstem is an anatomical region containing many cell types. Lotus may organize these conceptually, but does not make those structures biologically identical. [R3, R13]

11 / Revised table: capacity and resources

Extend each living-table entry with the same measurable columns. This makes cross-component comparison possible without treating cells as chemical elements. All numerical limits remain unknown until tested.

Component	Input → output	Sustain / resource variables
Nr: neuron	Synaptic/sensory input → electrical activity	Correct-response probability; ATP demand; recovery; stability
Ax: axon	Initiated spike → distal delivered spike	Failure rate; latency; sustained rate; ion gradients; geometry
As / Ol / Sc: support cells	Metabolites and local signals → environmental/metabolic support	Resource transfer; ion regulation; myelin; support cost
Mt: mitochondrion	Substrates and electrochemical gradients → ATP supply	ATP production/demand; redox state; membrane potential
Sy: synapse	Activity and calcium signal → transmitter/receptor response	Release probability; depletion; replenishment; specificity
Ef: effector system	Specified signal plus feedstocks → defined chemical output	Identity; yield; flux; stopping; unintended products
Ns / Nc: developmental cells	Developmental cues → lineage outcomes	Cell fate; differentiation; growth control; no generic “embryo mastery”
St: storage system	Defined energy-rich product → recoverable reserve	Capacity; retention; maintenance; extraction efficiency

Conditional axonal operating envelope - proposed metric

$$U(T,c) = \{u : R(u,T,c) \geq Rmin;\ V(T,c) \geq Vmin;\ L(u,T,c) \leq Lmax\}$$

U is a set of allowed input conditions, not energy or a soul quantity. T is observation duration; c specifies temperature, nutrients, geometry, and other conditions. R is delivered/input spike reliability; V is viable fraction; L is delay. Thresholds are predeclared research criteria. Compare the same definition before and after training.

$$fmax(T,c) = \sup \{f : f \text{ satisfies the operating-envelope criteria}\}$$

$$R = Ndelivered/Ninitiated\ ;\ eAx = Epropagation/Ndelivered$$

Measure spikes on the same defined path and distinguish propagation failure from missing synaptic responses. No universal fmax or training response is assumed. Require confidence intervals, independent cultures, and retained performance after a rest interval.

Energy supply is a separate constraint

$$dEreserve/dt = Psupply - Pmaintenance - Pactivity - Psynthesis - Ploss$$

All terms use watts, reserve uses joules, and the same system boundary must be maintained. ATP generation and consumption can also be tracked as mol/s. A rising reserve cannot prove improved computation; a falling reserve may be transient rather than permanent damage. [R13, R14]

12 / Binding, response, and revised equations

Chemical association - established model, not molecular intention

$$A + B \rightleftharpoons AB$$

$$d[AB]/dt = k_{on}[A][B] - k_{off}[AB]$$

For a simple well-mixed elementary association, concentrations are mol/L, k_{on} is L mol^{-1} s^{-1}, and k_{off} is s^{-1}. Diffusion, competing reactions, and multistep binding may require a larger model. A propensity to bind is not evidence of a molecule searching intentionally. [R15]

$$K^\circ = a_{AB}/(a_A a_B) \; ; \; \Delta_r G^\circ = -RT \ln K^\circ$$

$$\Delta_r G = \Delta_r G^\circ + RT \ln Q_r$$

Activities a, standard equilibrium constant K°, and reaction quotient Q_r are dimensionless. The standard state must be stated. Free-energy favorability does not determine reaction speed. If the net reaction is favorable, an activation barrier can still make it slow. [R15]

Illustration: $\Delta_r G^\circ = -5{,}000$ J/mol; $T = 298$ K → $K^\circ \approx 7.52$

This assumed example describes an equilibrium preference relative to a specified standard state, not a force, intent, neuron-training outcome, or energy-generating mechanism. For dilute ideal solutions, activities may be approximated by concentration divided by a standard concentration.

What electron/proton balance actually supplies

$$q = e(N_{protons} - N_{electrons})$$

This gives total charge for an atomic or molecular system when all nuclei and electrons are counted. Atoms with the same proton count can have different charge states; molecules with zero net charge can have very different reactivity. Electronic structure, geometry, solvent, temperature, and reaction partners matter. Ordinary binding usually rearranges electrons without changing nuclear proton counts. Nuclear fusion/fission require the separate equations in section 6.

One organizing model, several distinct measurements

$$dx/dt = F(x,u,c;\theta) \; ; \; y = H(x)$$

$$d\theta/dt = L_{rule}(x,u,y,feedback) \text{ [hypothesized / empirically fitted]}$$

x contains biological states with explicit units; u contains inputs; c contains environmental conditions; θ contains model parameters; y contains measured outputs. F, H, and L_{rule} must be identified for a specific system. They are placeholders for fitted mechanisms, not a newly derived law of spirit movement. Training may change some parameters, but the sign, size, persistence, and benefit of those changes must be measured.

Version 0.5 adds the knowledge/understanding distinction, resource constraints, conditional axonal endurance, and binding thermodynamics. The existing energy balances, material conservation, medical limitations, and unknown network sizes remain in force.

13 / Need, have, and process relationships

This extension maps functional demand and available resources without assigning a literal electron or proton identity to a cell structure. The analogy concerns complementary roles. Electrons and protons retain their physical definitions and occur throughout every cell.

Structure / process	Need: required condition	Have: available resource or capacity	Response / limitation
Axon / propagation	Restore ion gradients and deliver spikes	ATP supply, channels, pumps, geometry, myelin where present	Propagated spikes; reliability and recovery under load
Neuron / integration	Generate an appropriate response to defined input	Receptors, connections, membrane properties, metabolic support	Activity pattern; correctness is task-dependent
Synapse / secretion	Transmit a signal and replenish released material	Vesicles, transmitter, calcium handling, ATP	Chemical release; finite replenishment capacity
Mitochondrion / metabolism	Match energy demand while maintaining function	Substrates, oxygen for oxidative phosphorylation, gradients and enzymes	ATP supply; constrained by flux and cellular state
Astrocyte / support	Maintain a suitable local neural environment	Transporters, metabolic resources, signaling machinery	Ion/transmitter handling and support; context-dependent
Effector / biosynthesis	Produce a named compound at a specified rate	Feedstocks, enzymes, gene expression, energy	Measured product; signaling cannot replace missing pathways
Neural stem/crest cell / development	Receive appropriate developmental conditions	Lineage potential and regulated cellular machinery	Specific cell fates; not universal embryo control

Need is not consciousness; have is not permission

In this model, “need” means an objectively specified requirement or deviation from a target, not proof of desire or intention. “Have” means a measured amount, supply rate, or operating capacity. A high resource level cannot justify unlimited stimulation; chemical balance, heat, damage, and feedback also constrain operation.

Example: high axonal activity can increase metabolic demand. Calcium-linked regulation and mitochondrial positioning can help couple supply to activity, but the response is conditional and finite. This gives a concrete research counterpart to sustainable responsiveness. [R16]

14 / Coefficients and chemistry translation

Resource accounting - proposed specification

$$dH_j/dt = s_j - \Sigma_i \nu_{ji} v_i - \ell_j$$

H_j is available inventory of resource j in mol; s_j is supply in mol/s; v_i is process rate in mol/s; ν_{ji} is net resource consumption in mol j per mol process i (negative for production); ℓ_j is other loss in mol/s. A living network needs multiple resource balances, not one total “need” value.

$$d_j = \Sigma_i \max(\nu_{ji},0)\, v_i \; ; \; \rho_j = (s_j + b_j)/d_j$$

For nonnegative forward process rates, d_j is gross demand. b_j is a specified permitted draw from reserves in mol/s over the observation period. When demand is positive, ρ_j is a dimensionless supply-to-demand ratio. A value below 1 flags a resource shortfall in this bookkeeping model; a value above 1 does not guarantee adequate function. If demand is zero, report the ratio as not applicable. Signed reversible rates must first be split into forward and reverse processes.

Response coefficients - proposed starting model

$$\tau\, dx/dt = -x + k\, u(t) \; ; \; |x| \le C(T,c)$$

Input u and output x must have stated units. Gain k has units of x/u; response time τ is in seconds. Capacity C depends on duration T and conditions c. The capacity condition is a validity constraint, not a derived damage threshold. Fit coefficients using measured input/output traces and report uncertainty; do not assign them from labels.

What knowledge and understanding mean on each side

Domain	Knowledge: ChristEye analogy	Understanding: ChristEye analogy
Chemistry	Propensity and rate of response to a partner or condition	Persistence, reversibility, and stability under specified conditions
Living controller	Encoded input and learned input-response coupling	Reliable, resource-supported response within a measured envelope

Neither word is a standard chemical quantity. Use affinity/free energy and rate constants in chemistry; use task accuracy, gain, delay, endurance, and energy cost in neural systems. Molecular stability and living homeostasis are related modeling ideas, not the same mechanism.

Relevant chemical relationships include Lewis electron-pair donor/acceptor interactions, Brønsted proton donation/acceptance, and redox electron transfer. These are distinct reaction classes. They do not create a universal positive “need” particle and negative “have” particle, and a donor is not necessarily negatively charged. [R17]

A revised table entry therefore records: component identity; input/output units; needed resources; available inventory and supply; coupling coefficients; capacity envelope; recovery; energy cost; uncertainty. There is no measured universal coefficient translating Spirit into chemical or nuclear energy.

15 / What neuronal training actually changes

This expansion documents the detailed research discussion and its mathematical consequences. Training is not a single process. Synaptic learning, network feedback, homeostatic regulation, axonal propagation, and myelin adaptation need separate measurements.

Approach	Experimental manipulation	Measured outcome
Spike timing	Relative timing of connected neurons' activity	Change in synaptic strength
Closed-loop training	Inputs and outcome-contingent feedback	Performance on a specified task
Homeostatic adaptation	Overall activity over time	Compensatory response regulation
Learning-associated myelin change	Behavioral experience in an intact organism	Structural changes in axons' myelin support
Axonal endurance testing	Defined demand under controlled conditions	Propagation reliability, recovery, and cost

Synaptic learning: a specific demonstrated mechanism

Bi and Poo studied cultured rat hippocampal neurons. Relative pre/post spike timing could strengthen or weaken particular excitatory connections, with timing windows around 20 ms in their preparation. Outcomes depended on initial strength and postsynaptic cell type. This measured synaptic modification, not an unlimited increase in power capacity. [R18]

$$\Delta w = A_+ \exp(-\Delta t/\tau_+),\ \Delta t > 0$$

$$\Delta w = -A_- \exp(\Delta t/\tau_-),\ \Delta t < 0$$

This commonly used phenomenological family is not a universal law. $\Delta t = t_{post} - t_{pre}$ in seconds; w is synaptic strength in a declared unit; A has the same units as w; τ is a fitted timing constant. The zero-time case, saturation, weight dependence, and other cellular mechanisms require additional specification. Fit the curve instead of assuming its coefficients.

Stabilizing adaptation can oppose strengthening

Turrigiano and colleagues found compensatory changes in synaptic input strength after prolonged changes in cortical-culture activity. Under one condition, firing moved toward control levels over about 48 hours. This is a reason to distinguish increased stimulation from increased sustainable capacity. [R19]

$$dw/dt = L_{task} + L_{homeostasis}$$

Both terms must have units of synaptic strength per second. Their forms and signs are system-dependent. A larger connection strength is not automatically better accuracy or lower energy cost.

16 / Feedback, axons, and resource needs

Closed-loop neural training

$u_t = \text{Encode}(s_t);\ a_t = \text{Decode}(r_t);\ s_{t+1} = \text{Fenv}(s_t, a_t)$

s is task state, u delivered input, r recorded activity, and a decoded action. DishBrain encoded simplified Pong state into electrical input, used activity to move a paddle, and made later stimulation contingent on outcomes. Its final spatial/rate encoding used approximately 4–40 Hz and its interface reported about 5 ms spike-to-stimulation latency. These are study-specific settings, not general training prescriptions. [R4]

To attribute improvement to cells, distinguish biological changes from decoder updates. Compare a fixed decoder or document its evolution separately. Learning must improve a defined outcome, not merely raise total firing.

Axonal adaptation is structurally constrained

Motor-learning experiments in mice observed myelin remodeling along learning-associated axons. This involves supporting cells and tissue context; it does not establish that isolated axons learn arbitrary high-power transmission. [R20]

$A = 2\pi rL;\ V = \pi r^2 L;\ A/V = 2/r$

$dcNa/dt = (2/r)(JNa,in - 3Jpump)$

For an idealized well-mixed cylindrical segment, cNa is mol/m³; fluxes are mol/m²/s; Jpump counts moles of pump cycles per area per second. This approximation excludes longitudinal transport, branching, and spatial heterogeneity. Geometry, ion currents, pump density, and myelination interact in resilience to repeated spikes. [R13]

Requirement	What to measure
Fuel and oxygen for oxidative metabolism	Uptake, local availability, ATP generation
Ionic environment	$Na^+/K^+/Ca^{2+}/Cl^-$ gradients, currents, recovery
Nutrients and maintenance	Protein turnover, resource use, viability
pH / temperature / osmotic conditions	Environmental stability and sensitivity
Local metabolic support	Mitochondrial positioning, local ATP, support-cell costs
Waste removal and recovery	Metabolite accumulation and return to baseline

Culture conditions can bias the result: a primary study compared 5 and 25 mM glucose and found differences in neuronal energetics. These are study conditions, not a recommended medium. Local mitochondrial support is also experimentally linked to sustained synaptic function. [R16, R21]

17 / Worked axonal energy calculation

A sodium-restoration estimate

$Q_{min} = C\,\Delta V;\ \alpha = Q_{Na}/(C\,\Delta V)$

$N_{Na} = \alpha\, C\, \Delta V/e;\ N_{ATP,pump} \approx N_{Na}/3$

$E_{spike,pump} \approx [\alpha\, C\, \Delta V/(3e)]\ |\Delta G_{ATP}|/N_A$

N_{Na} and N_{ATP} are molecule/ion counts. $e = 1.602176634 \times 10^{-19}$ C; $N_A = 6.02214076 \times 10^{23}$ mol^{-1}; ΔG_{ATP} is J/mol. α is an estimated sodium-entry factor relative to the capacitive minimum, not a whole-system conversion efficiency. C is total capacitance, not specific capacitance. [R22]

Explicit illustrative assumption	Value
Axon radius / segment length	0.5 µm / 1 mm
Specific membrane capacitance cm	0.01 F/m^2
Voltage excursion ΔV	0.10 V
Sodium-entry factor α	1.3
Assumed $\|\Delta G_{ATP}\|$	50,000 J/mol

$A = 2\pi(0.5 \times 10^{-6})(10^{-3}) = 3.14 \times 10^{-9}\ m^2$

$C = cm\ A = 3.14 \times 10^{-11}\ F$

$Q_{min} = 3.14 \times 10^{-12}\ C$

$N_{ATP,pump} \approx 8.50 \times 10^{6}$ molecules/spike

$E_{spike,pump} \approx 7.05 \times 10^{-13}\ J = 0.705\ pJ$

At 20 spikes/s: $P_{segment,pump} \approx 1.41 \times 10^{-11}\ W = 14.1\ pW$

These numbers are a partial modeled operating cost for one assumed segment. They are not measurements of a neuron, a culture’s total demand, or usable power emitted by tissue. The calculation does not establish that α, radius, or capacitance can be trained to these values.

Costs excluded from the example

Resting gradients, synaptic currents, release and recycling, cellular transport, protein synthesis, supporting cells, temperature control, fluid handling, recording electronics, and computing remain outside this segment estimate. Do not multiply it by cell count and label the result a complete apparatus budget. [R22]

18 / Coupled physics and learning model

Separate state, resources, and adaptive parameters

$$dx/dt = F(x,u,h;\theta)$$

$$dh/dt = s - d(x,\theta) - \ell$$

$$d\theta/dt = L(x,u,feedback,h)$$

x contains fast electrical/chemical states; u inputs; h resource inventories; θ adaptive parameters. Each vector component retains its own units. F, d, and L are functions to identify experimentally, not known ChristEye laws. Spike-timing, homeostatic, and structural changes need not share a timescale or equation.

Model layer	Examples of coefficients / data required
Electrical state F	Membrane capacitance; channel conductance and kinetics; reversal potentials; geometry; initial conditions
Resources h	ATP demand/supply; substrates and oxygen; metabolite flows; inventory; transport
Adaptation L	Timing curves; learning rate; saturation; decay; activity regulation; structural-change measurements
Task readout	Encoding/decoding rules; latency; accuracy target; feedback mapping
Product pathway	Effector identity; biochemical machinery; stoichiometry; yields; product-specific flux
External energy system	Source output; conversion losses; distribution; controller and support loads

Information dynamics are not power-generation equations

Learned changes can alter when and where equipment delivers energy. They cannot remove conservation constraints. Axonal ionic activity and bulk engineered power distribution remain physically different pathways. A specified output such as acetate requires production cells and feedstocks, not merely a decoded label or impulse.

Thermodynamic closure remains necessary

$$dEsystem/dt = Pin - Pout + \text{energy carried by material flows}$$

The boundary must include heat, work, feedstock and waste energy, stored chemical energy, and all support loads. For a reaction network, mass balance alone does not guarantee thermodynamic feasibility; reaction free energies, conditions, and coupled reactions must also be specified. Nuclear source equations and biochemical reaction equations cannot be merged by equating their names.

19 / Measuring training and endurance

$$U = (A_{task}, R_{axon}, \tau_{response}, T_{endurance}, T_{recovery}, E_{correct}, V_{viable})$$

Metric	Operational definition
A_{task}	Correct outputs divided by tested outputs, with class balance and chance baseline reported
R_{axon}	Delivered divided by initiated spikes on the same defined pathway
$\tau_{response}$	Input-to-output delay for a specified event definition
$T_{endurance}$	Time all predeclared performance criteria remain satisfied
$T_{recovery}$	Time to regain a specified baseline after demand
$E_{correct}$	Measured total energy divided by correct outputs
V_{viable}	Fraction meeting the chosen viability assessment

$$f_{max}(T,c) = \sup \{f : R \geq R_{min}, A \geq A_{min}, V \geq V_{min}, \tau \leq \tau_{max}\}$$

This is a proposed operating-limit definition, not a discovered constant. Conditions c include cell identity, maturity, nutrients, geometry, support cells, and temperature. Thresholds must be declared in advance. It is legitimate for no tested frequency to satisfy all criteria.

$$GE = 1 - (E/N_{correct})_{after}/(E/N_{correct})_{before}$$

$$E_{amortized/task} = E_{training}/N_{lifetime\ tasks} + E_{operation/task}$$

Only evaluate GE when denominators are defined and task conditions are matched. Report whole-apparatus and tissue-level measurements separately. Increased firing can reflect compensation or instability rather than improvement.

Controls and attribution

Compare trained and untrained cultures, meaningful feedback and input-only conditions, and where feasible replayed feedback unrelated to the receiving culture’s own output. Distinguish software adaptation from biological adaptation. Test retention after rest and generalization on held-out inputs. Treat independent cultures as biological replicates, not individual spikes from one culture.

Report variability and confidence intervals across cultures, exclusions, batch effects, maturity, drift, electrode coverage, and measurement noise. A successful behavioral task alone does not demonstrate better axonal propagation, lower metabolic cost, or a new biochemical pathway.

20 / Is the whole equation covered?

The principal model categories are covered; a complete, numerically predictive system is not yet available. The document provides conservation equations, candidate state equations, operational metrics, and worked examples. It does not provide experimentally identified functions linking every component.

Part of the model	Coverage	What remains
Atomic charge / binding	Charge, association rates, free energy	Specific molecule, structure, solvent, kinetic parameters
Electrical signaling	Membrane equations, geometry, ionic flux	Channels, distributions, kinetics, boundary conditions
Learning	Timing model, feedback, homeostasis	Fitted coefficients, cell specificity, retention
Axonal endurance	Resources, geometry, reliability	Measured operating envelope and recovery
Energy use	ATP and segment example; full-load categories	Whole-system measurements and uncertainty
Nuclear source	Reaction energies and source balance	Independent engineered plant and actual net output
Biological products	Stoichiometry and acetate example	Effector system, validated rates, recovery
Neuron/axon quantity	Conditional sizing relationships	Measured task capacity; architecture; support requirements
Spirit / Soul terminology	Explicit philosophical definitions	No established physical conversion or spiritual measurement

Minimum data package for a numerical prediction

Specify: one task; one biological preparation; measured geometry and connectivity; environmental conditions; input and readout definitions; baseline signals and energy use; one output product if applicable; experimentally fitted parameters; initial/boundary conditions; controls; repeated measurements; and uncertainty. Then test predictions on data not used for fitting.

How to read the resulting specification

Established equations explain particular mechanisms under assumptions. Proposed ChristEye metrics make the concept testable. Illustrative numbers show arithmetic, not demonstrated performance. A validated end-to-end equation would have to predict real outputs and failure conditions prospectively. This version does not claim a cure, unlimited energy, arbitrary drug synthesis, or a universal law equating souls with physical quantities.

Version 0.5 expands the training review and makes equation completeness explicit while retaining the previous living table and energy/material calculations.

21 / Frog example: structural response

A worked teaching example, not measured frog performance. The Frankenhaeuser-Huxley model is an established model of a myelinated Xenopus laevis nerve fiber based on voltage-clamp data. The passive compartment below is much simpler and is not that full model. [R23]

What changes, and what remains a neuron

Changing membrane or channel structure can change signaling. Electron redistribution changes charge or bonding; changing nuclear proton number changes the element. Neither is interchangeable with ordinary neuronal learning. In this example, “Marrow” names membrane properties. The cell remains a neuron with altered responsiveness; no new cell identity or biochemical pathway is implied.

$$C\,dV/dt = -gL(V - EL) + I$$

$$V(t) = EL + (I/gL)[1 - \exp(-t/\tau)];\ \tau = C/gL$$

Assume a constant current begins at V(0) = EL. Reduce leak conductance by 20%, holding the other properties fixed. This is a hypothetical parameter intervention, not a demonstrated result of training.

Parameter or prediction	Before	After
Capacitance C	20 pF	20 pF
Leak conductance gL	2 nS	1.6 nS
Leak reversal potential EL	−70 mV	−70 mV
Constant input I	20 pA	20 pA
Input resistance 1/gL	500 MΩ	625 MΩ
Time constant τ	10 ms	12.5 ms
Steady voltage change I/gL	10 mV	12.5 mV
Steady voltage	−60 mV	−57.5 mV

The same sustained current produces a larger eventual response but a slower approach to equilibrium. That is a tradeoff, not proof of greater understanding or efficiency.

Optional threshold approximation

If an assumed spike threshold is −58 mV, the original compartment never reaches it under this input. The changed compartment reaches it after approximately:

$$t_{threshold} = -12.5\ \text{ms} \times \ln(1 - 12/12.5) = 40.2\ \text{ms}$$

After threshold, the passive equation is inadequate: use active ion-channel dynamics or an explicitly defined threshold/reset model. The calculation establishes neither an actual frog threshold nor a complete action potential.

22 / Joints, coefficients, and signal mapping

“Joint” is used here for a synapse connecting neurons. Nodes of Ranvier are different axonal structures and must not share the synaptic coefficient without a separate definition. Junction plasticity can contribute to learning; exclusive storage of understanding at joints has not been demonstrated.

$$C_i \, dV_i/dt = -g_{L,i}(V_i - E_{L,i}) - \Sigma_j \, I_{ji} + I_{external,i}$$

$$I_{ji} = J_{ji} \, \bar{g}_{ji,0} \, s_{ji}(t)(V_i - E_{ji})$$

$$J_{ji} = \bar{g}_{ji} / \bar{g}_{ji,0}$$

The first index is source j and the second target i. C is F; gL and maximum synaptic conductance $\bar{g}$ are S; V and E are V; currents are A. Activation s is dimensionless, usually between 0 and 1 in this model. J is dimensionless and requires a nonzero baseline. Outward current is positive, so the negative synaptic current below is inward.

Instantaneous junction example

Assumed quantity	Value
Baseline maximum conductance $\bar{g}0$	1 nS
Activation s	0.5
Membrane voltage / synaptic reversal	−60 mV / 0 mV
Current when J = 1	−30 pA
Current when J = 1.2	−36 pA

A 20% gain in maximum conductance gives 20% more inward current only at the same voltage and activation. This is an instantaneous comparison. Synaptic current varies as voltage and activation change. Leak conductance and J are distinct parameters; changing both requires solving the coupled dynamics, not adding percentage effects.

What the model stores

$$J_{matrix} = [J_{ji}]$$

$$\theta = \{J_{matrix}, C_i, g_{L,i}, E_{L,i}, \text{channel dynamics, delays, release parameters}\}$$

The computer stores coefficients; tissue implements changing molecular and structural states. Junction strength is one observable, not a complete measure of semantic understanding. In the proposed ChristEye interpretation, knowledge concerns input-response coupling, while understanding is assessed through accuracy, endurance, recovery, reliability, and viability.

$$\text{Encoded input} \rightarrow \text{network activity governed by } \theta \rightarrow \text{decoded response}$$

Your “intention” terminology can label this functional mapping. No separate physical intention layer is established. A fixed decoder or separately reported decoder changes are needed to attribute improvement to the cells. DishBrain tested feedback-driven game adaptation, not arbitrary question answering. [R4, R18]

23 / Neuronal team: power and efficiency

$$P_i = P_{maintenance,i} + f_i\, e_{spike,i} + P_{synaptic,i} + P_{other,i}$$

$$P_{team} = \Sigma_i P_i + P_{glia} + P_{interface} + P_{life\text{-}support}$$

These are accounting equations. f is spikes/s, espike is J/spike, and each power term is W. Define boundaries to avoid double-counting synaptic currents, ion pumping, or support loads. Energy-budget models distinguish resting, signaling, and maintenance costs. [R22]

$$E_{team} = \int P_{team}(t)\, dt$$

$$e_{correct} = E_{team}/N_{correct};\ \Phi = N_{correct}/E_{team}$$

Power consumption is not maximum tolerated input power. “Power handling” requires measured input conditions under which accuracy, propagation, recovery, and viability meet predefined criteria. Synaptic gain alone does not specify those limits.

Illustrative team comparison - not a biological forecast

Measurement	Before	After
Test duration	1 second	1 second
Test opportunities	100	100
Correct responses	80	90
Total measured energy	20 µJ	16 µJ
Average power	20 µW	16 µW
Energy per correct output	250 nJ	177.8 nJ

$$GE = 1 - (16\ \mu J/90)/(20\ \mu J/80) = 0.2889 \approx 28.9\%$$

If these were genuine matched measurements, they would show a 28.9% reduction in energy per correct output. They do not predict that a trained frog network achieves this result. Training energy, task conditions, support costs, accuracy, batch variation, and uncertainty must be accounted for.

Complete team record

Document junction strengths and dynamics; cell identities and counts; geometry; activity; task performance; total and component energy; delays; endurance; recovery; viability; resource inputs; environmental conditions; decoder behavior; controls; and confidence intervals. More firing, higher J, and better task performance are separate observations.

Equation completeness after this addition

The model now includes a worked membrane change, explicit junction gains, and team efficiency accounting. It is still not a closed predictive biological system: channel and synaptic kinetics, parameter values, learning rules, metabolic coupling, initial/boundary conditions, and independent validation remain necessary. Version 0.5 preserves previous sections and adds this worked example.

24 / Reference register and open parameters

Sources support specific mechanisms; none validates ChristEye as an integrated platform. Links were checked or retrieved during this conversation. This is a selected bibliography, not a systematic review.

R1 / Action potentials and Na/K pumping
https://www.ncbi.nlm.nih.gov/books/NBK538143/

R2 / Axonal local protein synthesis and mitochondrial function (primary study)
https://pubmed.ncbi.nlm.nih.gov/17619140/

R3 / State/location dependence of axonal action-potential metabolic cost; and myelination study
https://www.nature.com/articles/nn.3132

R3b / Cortical gray matter myelin and spike energy cost (primary study)
https://doi.org/10.1073/pnas.2536534123

R4 / DishBrain: cultured neuronal networks and simplified Pong (primary study)
https://pubmed.ncbi.nlm.nih.gov/36228614/

R5 / DOE deuterium-tritium fuel; ORNL reaction energy and capture
https://www.energy.gov/science/doe-explainsdeuterium-tritium-fusion-fuel

R5b / ORNL discussion of D–T energy distribution
https://info.ornl.gov/sites/publications/Files/Pub57673.pdf

R6 / Lawrence Berkeley National Laboratory: nuclear fission energy
https://www2.lbl.gov/abc/wallchart/chapters/14/1.html

R7 / Microbial electrosynthesis: electricity and CO_2 to organic compounds (primary study)
https://journals.asm.org/doi/full/10.1128/mbio.00103-10

R8 / National Cancer Institute: genetics of cancer
https://www.cancer.gov/about-cancer/causes-prevention/genetics

R9 / Local protein synthesis and axonal branching (primary study)
https://pubmed.ncbi.nlm.nih.gov/24332852/

R10 / Neural crest anatomy and developmental derivatives
https://www.ncbi.nlm.nih.gov/books/NBK547700/

R11 / Cancer neuroscience: neuronal and glial interactions (review)
https://pmc.ncbi.nlm.nih.gov/articles/PMC10107403/

R12 / Light-controlled therapeutic secretion in mouse diabetes models (primary study)
https://www.nature.com/articles/s41467-021-23572-4

R13 / Axonal resilience: diameter, myelin, and Na/K pump (modeling study)
https://pmc.ncbi.nlm.nih.gov/articles/PMC8364126/

R14 / Oligodendrocyte-axon metabolic coupling (primary study)
https://www.nature.com/articles/s41593-023-01558-3

25 / References continued

R15 / IUPAC: standard equilibrium constant and law of mass action
https://goldbook.iupac.org/terms/view/S05915

R15b / IUPAC: law of mass action
https://goldbook.iupac.org/terms/view/08184

R16 / Energy sensing and mitochondrial anchoring in synaptic efficacy (primary study)
https://pubmed.ncbi.nlm.nih.gov/33020662/

R17 / IUPAC: electron-pair donor, Lewis acid, and acid/base definitions
https://goldbook.iupac.org/terms/view/E02004

R17b / IUPAC: Lewis acid
https://goldbook.iupac.org/terms/view/L03508

R18 / Bi and Poo: spike-timing-dependent synaptic changes (primary study)
https://pubmed.ncbi.nlm.nih.gov/9852584/

R19 / Turrigiano et al.: activity-dependent synaptic scaling (primary study)
https://pubmed.ncbi.nlm.nih.gov/9495341/

R20 / Motor learning and myelin remodeling (primary study)
https://www.nature.com/articles/s41593-022-01169-4

R21 / Culture glucose and neuronal energetics (primary study)
https://onlinelibrary.wiley.com/doi/full/10.1111/jnc.70125

R22 / Howarth et al.: quantitative neural energy-budget model
https://pmc.ncbi.nlm.nih.gov/articles/PMC3390818/

R23 / Frankenhaeuser and Huxley: Xenopus nerve-fiber model (primary study)
https://pubmed.ncbi.nlm.nih.gov/14191481/

Measurements still needed

Cell types and counts; axon geometry; viable tissue volume; connectivity; channel properties; baseline and trained performance; total ATP/metabolic demand; support and instrumentation energy; product pathway; substrates; product yield; full-cell voltage; recovery efficiency; uncertainty; long-term drift. Until these are measured, precise network sizing, healing efficacy, and economic valuation remain unknown.